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Palaeomagnetic Stratigraphy of Tertiary Lavas from Northern Ireland

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Summary

Three lava profiles, sampled at 1·3 km and 2·3 km spacings, overlap stratigraphically as proven by two geological marker horizons. Groups of several successive lavas often have remarkably coherent directions of magnetization. Further, on three occasions, a group of lavas in one profile correlates palaeomagnetically with a group in another profile, and these correlations agree very well with the geological marker horizons. A lava-mean α_{95} of 3·1° makes possible palaeomagnetic stratigraphy based on secular variation, apparently within one polarity epoch (reversed).

Since many lavas seem to have recorded virtually the same magnetic field, 54 distinct lavas contribute only 25 distinct spot values of the ancient field, from which we deduce a mean pole position at 69·6° N and 162·9° E, with $P = 5°$ and $M = 7°$.

1. Introduction

As part of an overall detailed survey of the British Tertiary volcanic province, in April 1967 we sampled the longest known section of basalts in Northern Ireland. This contained nominally 58 lavas in three profiles (Fig. 1) from Garron Point on the East coast of County Antrim (lat. 55·05° N, long. 6·05° W). Fig. 1 shows that there is overlap between these sections.

The analysis in Section 2 suggests that sufficiently accurate methods and detailed sampling permit the use of natural remanent magnetism for stratigraphical correlation even without polarity inversions. The smaller within-epoch variations of the magnetic field can confer enough distinction on groups of lavas to make them useful as marker horizons.

The other pertinent point is that the finding of mean directions and mean pole positions can be deceptive, since even very large groups of lavas or dykes can be extruded over a short enough time that they do not average out the time effects (Section 3).

A few smaller studies are reported in Section 4.

2. Palaeomagnetic stratigraphy

Four drill-core samples, separately oriented using geographic sighting, were taken from each lava. Alternating field demagnetization on each core was done in 50 oersted steps from 0 to 200 oersteds. All the data were then plotted and reviewed for all

* Received in original form 1970 January 15

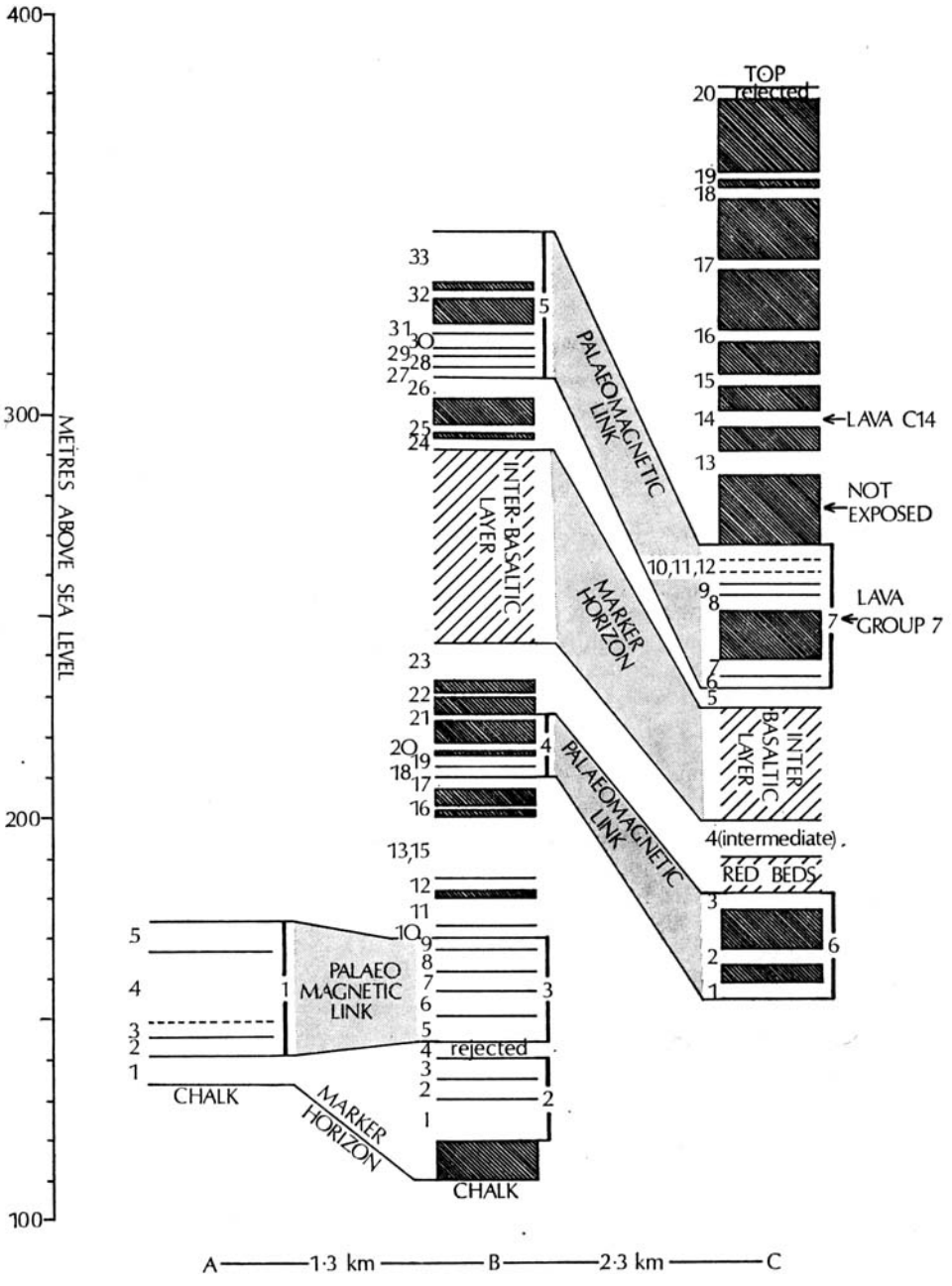


FIG. 1. Three lava profiles, A, B and C, from Garron Point in N. Ireland (55·05° N, 6·05° W).

A—just above chalk on north side of Cushenilt Burn (co-ordinates 3266 4244, Ordnance Survey of Ireland 1 in. Mid Antrim map).

B—a stream section 0·5 km due south of Milltown (stream not marked on 1 in. map) (co-ordinates 3253 4239).

C—a section following Allmore Burn up from where the trees cease, and ending on top of the plateau at 3242 4213.

An explanatory legend is on right of diagram. Also, a dashed horizontal line indicates uncertainty in the field about the reality of a lava boundary (see Section 4·3).

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cores at all alternating fields. Many cores were returned for further demagnetization and tracable errors were corrected. A few cores and two complete lavas were rejected due to extreme magnetic instability or to errors. All lavas were tilt corrected for the regional tilt at each profile as measured from a large distance by the author. Allowance

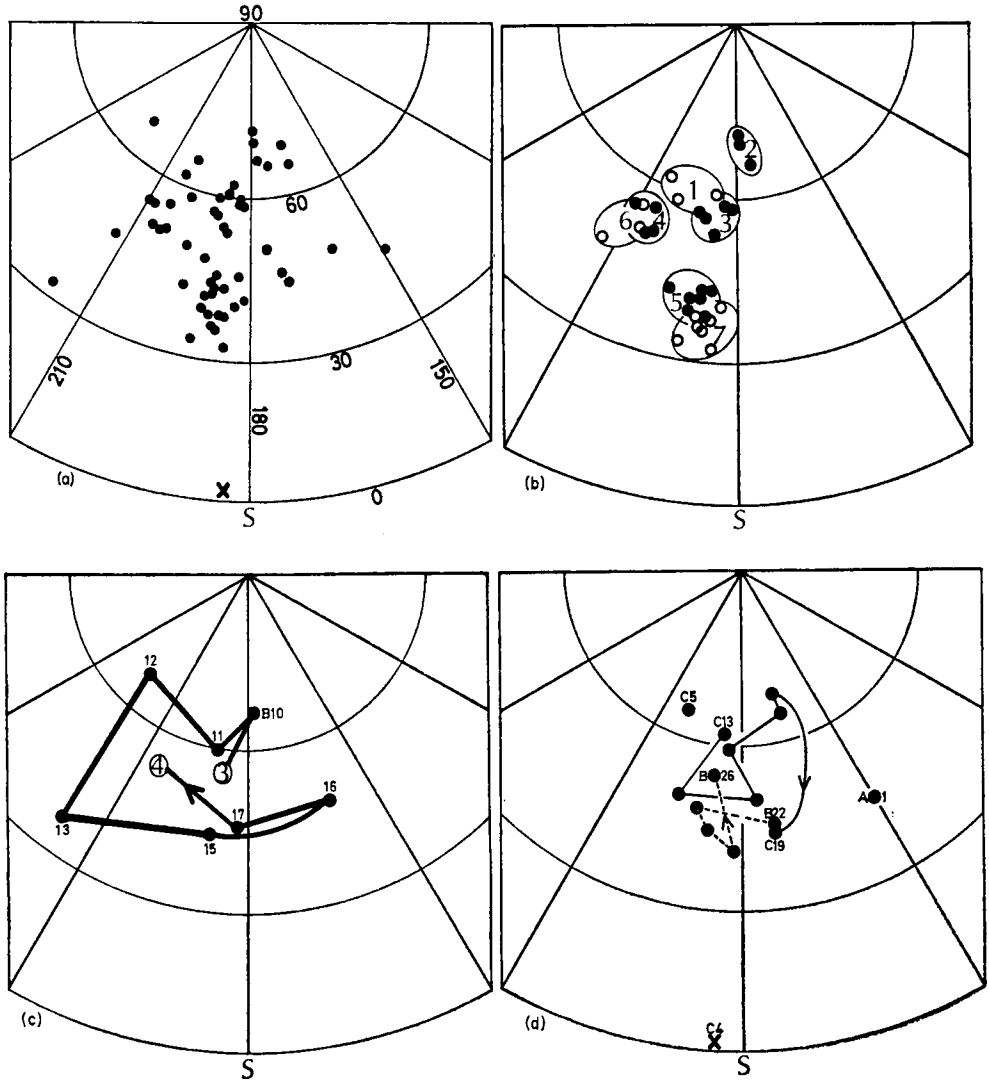


FIG. 2(a). The 54 lava-mean directions from Table 1. Note that the regions of high density are not accidental, but due to *consecutive* lavas with similar directions. These consecutive groups have been separately redrawn in Fig. 2b (see Section 2). Figs 2b+2c+2d are a breakdown of all of Fig. 2a. (b) Lava groups 1 to 7, each containing three or more consecutive lavas with closely similar directions or magnetization. Groups 1 and 3, 4 and 6, 5 and 7 agree with each other in pairs, which is the basis for the stratigraphic palaeomagnetic links drawn in Fig. 1. I consider that these 32 lavas define only four independent spot records of the ancient field. (c) Directions of magnetization for the lava succession B10 to B17, leading from palaeomagnetic group 3 to group 4. These lavas are discussed in Section 3 (concerning their independence in time) and in Section 4.1 (concerning the effects of faulting on remanence, and the non-existence of flow B14). (d) Directions of magnetization of the remaining lavas which appear to be independent from each other in time (Section 3). Lava C4 is the intermediate lava noted in Section 4.2.

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Table 1

Lava	N		Decl°	Incl°	α_{95}°	Remarks
A001	4		149·6	−44·7	2·8	
A002	3	(4)	202·9	−62·0	9·6	A2 to A5 are group 1.
A003	4		187·1	−60·3	2·2	A3 and A4 are the same
A004	4		186·8	−60·8	3·2	lava (Section 4.3)
A005	4		198·6	−58·6	1·9	
B001	4		179·2	−71·8	3·5	
B002	4		178·9	−69·6	2·0	B1 to B3 are group 3
B003	4		173·3	−65·4	4·1	
B003B	1		179·4	−67·6	—	Baked by B003
B004	4		—	—	—	Rejected
B005	3	(4)	183·5	−58·8	5·4	
B006	4		182·0	−58·5	2·8	
B007	4		186·1	−53·7	3·5	B5 to B9 are group 3
B008	4		190·9	−57·2	3·6	
B009	4		189·7	−56·6	2·9	
B010	4		177·4	−66·4	2·2	
B011	4		189·8	−59·7	1·9	
B012	3	(3 only)	224·1	−66·5	2·0	
B013	4	(originally 2)	216·9	−32·8	3·5	
B014	–	(originally 3)	—	—	—	Does not exist (Section 4.1)
B015	5	(originally 4)	188·7	−44·1	2·3	
B016	3	(4)	160·2	−48·6	1·8	
B017	4		182·5	−45·9	4·2	
B018	20		203·6	−56·0	1·5	
B019	4		209·5	−55·1	2·0	B18 to B21 are group 4
B020	4		203·0	−51·2	1·7	
B021	4		202·3	−51·5	3·0	
B022	4		172·8	−46·5	1·5	
B023	4		191·0	−48·7	1·9	
B024	3	(3 only)	187·8	−45·9	2·3	
B025	3	(3 only)	181·7	−41·5	1·7	
B026	4		187·4	−54·7	2·5	
B027	4		185·7	−43·5	5·9	
B028	4		189·4	−42·0	3·6	
B029	4		194·3	−43·1	3·0	
B030	4		188·0	−42·4	2·4	B27 to B33 are group 5
B031	4		186·2	−38·7	2·1	
B032	5		187·8	−43·3	2·2	
B033	6		198·8	−39·4	1·7	
C001	4		205·0	−50·9	3·2	
C002	4		212·2	−46·8	2·0	C1 to C3 are group 6
C003	4		207·6	−55·0	3·0	
C004	4		183·5	− 2·2	5·8	Intermediate direction
C005	4		200·9	−65·0	4·8	
C006	3	(4)	183·3	−40·6	7·1	
C007	2	(4)	185·9	−38·3	4·6	
C008	4		188·2	−38·5	2·5	
C009	4		187·5	−36·5	3·3	C6 to C12 are group 7
C010	3	(4)	190·8	−33·5	0·7	
C011	3	(4)	186·8	−35·9	3·1	
C012	4		184·9	−32·5	1·6	
C013	4		186·0	−62·1	2·8	
C014	4		195·9	−50·1	3·3	
C015	4		176·2	−50·8	2·2	
C016	4		184·1	−59·2	3·3	
C017	3	(4)	164·6	−65·0	7·4	
C018	4		166·3	−68·7	3·2	
C019	4		172·7	−44·9	3·0	
C020	6		—	—	—	Rejected

Lava-mean directions of magnetization for profiles *A*, *B* and *C*.
N = No. of independently oriented specimens used to find mean direction (numbers in parentheses indicate number actually collected where that differs from *N*).
α_{95} = 95 per cent of confidence for one lava.

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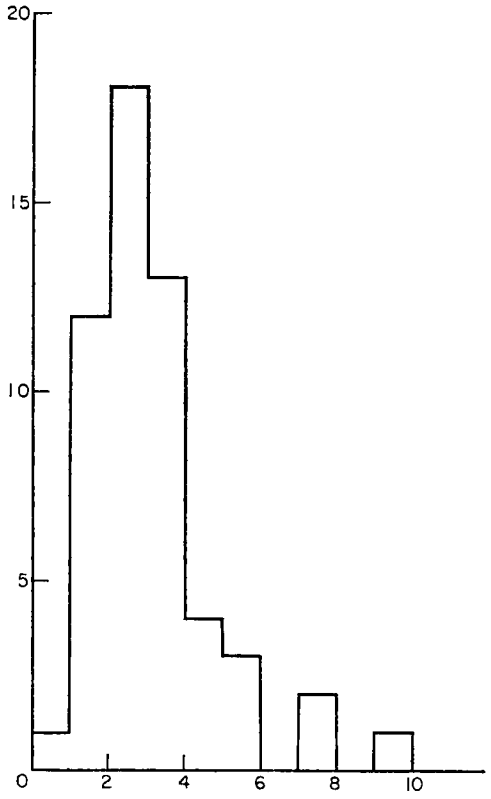


FIG. 3. A histogram of α_{95} for the 54 distinct lavas. The mean α_{95} is 3·1°. This level of accuracy seems adequate to permit palaeomagnetic correlation based on ancient secular variation within one polarity epoch.

was made for block faulting by measuring the tilt in the region of each of the three profiles.

The mean 95 per cent circle of confidence for a lava was 3·1°. Fig. 3 shows a histogram of α_{95}. It is clear that the errors are small compared with secular variations of a few tens of degrees, so that palaeomagnetic stratigraphy was in this case technically possible. Fig. 2(a) shows the mean direction of magnetization for each lava, as recorded in Table 1.

Fig. 2(b) and Fig. 1 show that there are seven groups of three or more consecutive lavas having closely equal directions of magnetization. Six of these groups agree closely in pairs (groups 1–3, 4–6, 5–7). Fig. 1 shows how these pairs of lava groups are linked stratigraphically. Two standard geological stratigraphic marker horizons are also available; the interbasaltic layer of highly laterized lavas (often *in situ*), and the chalk base of the lava pile, both of which are prominent and widespread in outcrop. Both of these marker horizons agree well with the palaeomagnetic links.

I interpret this as meaning that pairs of corresponding lava groups either were extruded almost simultaneously during lulls in magnetic field fluctuation, or were extruded so rapidly that the ancient field had no time to vary significantly during extrusion of the groups. One is perhaps seeing small time delays between extrusion of groups 5 and 7, for example. These palaeomagnetic groups probably correspond to G. P. L. Walker's ' flow-units ' extruded in rapid succession (Walker 1970) which have also been noted clearly in Iceland.

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If this is accepted, it means that the 32 lavas in Fig. 2(b) have recorded only four instantaneous values of the ancient field. Much of the information is therefore redundant as regards a time-average look at the ancient field, although not necessarily redundant for other uses.

Khan (1960) has used palaeomagnetic stratigraphy in Skye, on a lava-for-lava basis, rather than to correlate groups of lavas. The method of analysis has to be adapted to the evidence available in each instance. These Irish basalts do not exhibit a lava-for-lava correlation.

3. The mean direction of the ancient field

Of the 58 original lavas sampled, two (B4 and C20) were rejected because of palaeomagnetic scatter and two more lost their identity as described in Sections 4.1 and 4.3. Of the 54 remaining lavas, 32 are now thought to record only four instantaneous ancient field directions. What of the independence of the remaining 22 lavas? Figs. 2(c) and 2(d) show these 22 directions.

In Fig. 2(c), directions B10 to B17 in sequence are those from between lava groups 3 and 4. These may be independent lavas. There is not enough information to decide whether they represent a cycle of large secular variation between groups 3 and 4.

In Fig. 2(d), all directions are probably independent. Lavas B22 to B26 look somewhat grouped, but they straddle the interbasaltic layer, which often consists of up to four highly laterized lavas, and which must have taken considerable time to become laterized. Further, lavas C4 and C5 are in the lava interval corresponding to B22 to B26, and the C4 and C5 directions are nowhere near the B22–26 group.

Table 2

Group	No. of data	Mean declination	Mean inclination	α_{95}	Standard deviation
2	3 (lavas)	176·8°	−67·0°	—	2·8°
1 and 3	8	189·9	−58·4	3·1°	4·2
4 and 6	7	206·3	−52·4	3·0	3·7
5 and 7	14	187·7	−39·2	2·1	4·0

Overall mean direction (as in Section 3)

No. of data	Declination	Inclination	α_{95}	Standard deviation	R
25	184·7°	−54·3°	5·1°	13·8°	24·2786

Mean pole position

No. of data	Latitude	Longitude	P	M
25	69·6° North	162·9° East	5°	7°

Garron Point site Latitude = 55·05° North.
Garron Point site Longitude = 353·95° East.

Comparison with earlier work from Northern Ireland

No. of data	Declination	Inclination	α_{95}	Standard deviation	R	Authors
24 (igneous bodies)	194°	−60·2°	5·9°	14°	23·253	Hospers and Charlesworth (1954)
41 (igneous bodies)	183	−64·1	6·3	20	38·45	Wilson (1959)

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Therefore it seems that we may reasonably consider the 22 non-grouped lavas to be independent in time. Discarding the one intermediate lava, C4, the other 21 may be combined with the four directions associated with Fig. 2(b), giving the mean data in Table 2. Only one of the 25 lava mean directions exceeds the dipole field inclination (71·8° compared with 70·8°), so that there is certainly a real deviation from the present dipole field. Even so, there is some doubt about there being a simple interpretation of this mean. Every dyke and lava studied in Northern Ireland has been reversely magnetized (except for C4 here). With the exception of some Mull and Ayrshire dykes (Ade-Hall & Wilson 1969; Smith 1966; Raja 1964), this seems to be true of the rest of the British Tertiary province so far studied.

From this and other unpublished details, one gets the impression that the entire province may have been extruded in an extremely short time, short enough to accommodate not much more than one reversed epoch and perhaps less. If that turns out to be the case, then the whole of the British Tertiary province may not give good time-average data, but it would give enormously detailed information about a short period of time. There are about 2000 or more lavas (in profiles) and dykes (in swarms) easily available for study. We have now sampled about three quarters of these, and hope to resolve this problem.

4. Several subsidiary studies

A few particular studies were made, for example, to follow up questions entered into the field notes.

4.1 *The effects on natural remanent magnetization, of crushing in the close proximity of a fault*

The existence of the stream section of profile B is due to a fault, which begins as as few cm displacement at lava B6 (about 150 m altitude) and develops into a throw of more than 10 metres at lava B23. We kept to the western side of this fault in our collection, but initially worked close to it. Its effects became very pronounced until, after lava 15, we were forced to move onto the adjacent hillside to make sense of the stratigraphy.

For example, because of the crushing near the fault the distinction between B13, B14, and B15 could not be made by eye and this was noted in the field book. B13, B14 and B15 were found on palaeomagnetic grounds to be two lavas only; B14–1 and B14–2 belonging to lava B13, and B14–3 belonging to B15. This is why B14 is missing from Table 1, Fig. 1 and Fig. 2(c).

This result suggests that even when faulting has altered lavas to the extent that their boundaries are no longer visible, the natural magnetic remanence remains *coherent*. It does not necessarily mean that the remanence within a lava was not rotated (for example) by the shearing at the fault. The directions of B12 and B13 (Fig. 2(c)) are far enough from the rest of the directions to raise this suspicion but since we have not sampled B12 and B13 far away from the faulted zone, we cannot make further deductions.

4.2 *The intermediate flow C4*

This flow is isolated between a few metres of red beds below and the interbasaltic layer above. All four specimens had directions of magnetization which agreed well, and the specimens were sampled over a vertical range of three metres where there seemed no chance of block rotation of some of the lava. This is the only magnetically cleaned lava in the literature of the British Tertiary province, not having clearly reversed remanence (' cross ' in Fig. 2(d)).

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4.3 *Distinguishing successive lavas from each other*

Aside from the faulted region, there were three other places where the separate identity of lavas was not sure in the field. This was noted in the field books. Lavas A3 and A4 had lava mean directions as in Table 3 below.

Table 3

Lava	No. specs.	Mean declination	Mean inclination	α_{95}
A3	4	187·1	−60·3	2·2
A4	4	186·8	−60·8	3·2
C10	3	190·8	−33·5	0·7
C11	3	186·8	−35·9	3·1
C12	4	184·9	−32·5	1·6

Since their existence as separate lavas had been queried and their mean directions were extremely close (within 0·6°) it seems fairly certain they formed only one lava, and they were so treated.

Lavas C10, 11 and 12 were similarly uncertain. Lava C10 is clearly distinct from C11 and C12 at the 95 per cent level. C11 and C12 have overlapping 95 per cent confidence circles, but neither circle encloses the other mean. C10, 11, and 12 are the shallowest three directions in group 7 (Fig. 2(b)) where I have regarded them as distinct.

Lavas C13 to C19 were very similar petrologically, all being diktytaxitic and blocky. Since here near the plateau top there was much grass covering, their distinctness was uncertain. Their seven mean directions are shown in Fig. 2(d). The biggest α_{95} is 3·2° except for C17 (7·4°). They are all clearly distinct except for C17 and C18, which may be the same lava, although they are separated (vertically) by 7 or 8 m of unexposed region.

It can be seen therefore that natural remanence can often be used to decide about lava identities, but that there are borderline cases (C11, C12 and C17, C18) which cannot be resolved.

4.4 *A baked-rock study*

One baked specimen was collected from below lava B3.

Directions of magnetization are in Table 4.

Table 4

	Code	No. specimens	Declination	Inclination	α_{95}
Lava	B3	4	173·3	−65·4	4·1°
Baked rock	B3B	1	179·4	−67·6	—

The baked specimen agrees with the lava within 3·2° which lies within the 4·1° confidence circle for the lava.

5. Conclusions

The two major conclusions of this study are that (*a*) under favourable conditions of extrusion rates and magnetic stability, natural remanent magnetization can be

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used for purposes of stratigraphic correlation of lava groups. Ninety-five per cent confidence circles of 3° or 4° can be achieved on the mean direction for each lava; and this means that the scatter of all the lava-mean directions is due to ancient secular variation rather than experimental error; and (*b*) rapid extrusion rates relative to time variation of the ancient field can lead to much redundancy of palaeomagnetic data, to doubts about the error which should be associated with overall mean directions, and indeed about the propriety of the mean itself. The larger the number of lavas sampled in a formation or region, the more clearly it is seen that the ancient field has performed in very unsuspected ways, and the less confidence I personally have in reducing the data to a simple mean direction, or palaeomagnetic pole, with the usual statistical parameters. Doell (at the Madrid I.A.G.A. meeting, 1969) has expressed similar doubts about the significance of the error of an overall mean direction for all lavas sampled in the Hawaiian group of islands. However, at least an attempt can be made to allow for redundancy of data when finding a mean direction, as in Table 2 and Section 3.

Acknowledgments

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